

Causal Mediation Analysis: Natural Direct and Indirect Effects

Overview

Causal mediation analysis (CMA) addresses a fundamental scientific question: *how* does a treatment or exposure produce its effect on an outcome? Rather than merely establishing that A causes Y , mediation analysis decomposes the **total effect** into components that flow through an intermediate variable (the mediator M) versus those that bypass it. This module introduces the **potential outcomes framework** for mediation, defines **natural direct effects** (NDE) and **natural indirect effects** (NIE), and explores the critical **sequential ignorability** assumption required for identification.

We begin by critiquing traditional approaches (e.g., Baron & Kenny's product-of-coefficients method) and demonstrate how the counterfactual framework offers a model-free definition of mediation effects. Students will learn to work with **nested counterfactuals** (Y^{a,M^a}), understand the **composition assumption**, and apply **sequential ignorability** to derive identifiable estimands. The module integrates directed acyclic graphs (DAGs), hand calculations, and R implementations using `dagitty`, `ggdag`, and the `mediation` package.

By the end of this module, you will be equipped to critically evaluate mediation claims, recognize confounding threats specific to mediation analysis, and conduct principled causal mediation studies in R.

Learning Objectives

By the end of this module, students will be able to:

1. **Articulate** the limitations of traditional regression-based mediation approaches (Baron & Kenny) and explain why model-dependent definitions are problematic (30 minutes of study).

2. **Define** natural direct effects (NDE), natural indirect effects (NIE), and controlled direct effects (CDE) using nested counterfactual notation and interpret each quantity in plain language (45 minutes).
3. **State** the sequential ignorability assumption and identify which types of unmeasured confounding violate each component (20 minutes).
4. **Construct** DAGs representing mediation scenarios with potential confounders and determine adjustment sets for identifying natural effects (40 minutes).
5. **Execute** a complete causal mediation analysis in R, including data simulation, effect estimation, and sensitivity analysis (60 minutes of lab work).
6. **Distinguish** between NDE, NIE, and CDE conceptually and numerically through hand calculations and software output (30 minutes).

R Packages

```
# Load required packages
pacman::p_load(
  dagitty, # DAG construction and analysis
  ggdag, # DAG visualization with ggplot2
  mediation, # Causal mediation analysis
  tidyverse, # Data manipulation and visualization
  broom, # Tidy model outputs
  knitr, # Table formatting
  kableExtra, # Enhanced tables
  MASS, # Multivariate normal simulation
  MatchIt, # Propensity score matching (optional)
  latex2exp, # LaTeX expressions in plots
  patchwork # Combining plots
)

# Set seed for reproducibility
set.seed(579)
```

```
# Set default theme for plots
theme_set(theme_minimal(base_size = 12))
```

1 Comprehensive Topic Explanation

1.1 The Quest for Mechanisms: Why Mediation Matters

Scientific inquiry rarely stops at establishing *that* a treatment works. Researchers, policymakers, and clinicians want to understand *why* and *how* an intervention produces its effects. Does a literacy program improve children’s academic outcomes by increasing reading skills (M), by boosting self-confidence, or through some other pathway? Does a drug reduce mortality by lowering blood pressure, or through anti-inflammatory mechanisms? **Mediation analysis** provides a formal framework for answering these mechanistic questions (Hernán & Robins, 2020; Pearl, 2009).

A **mediator** M is an intermediate variable that lies on the causal pathway from exposure A (or treatment T) to outcome Y . The temporal ordering requirement is strict: $A \rightarrow M \rightarrow Y$. Mediation analysis decomposes the total causal effect of A on Y into:

- **Indirect effect:** The portion transmitted *through* M .
- **Direct effect:** The portion operating through *all other pathways*, excluding M .

Understanding mediation has practical implications. If the indirect effect dominates, interventions targeting the mediator may be effective. If the direct effect is large, alternative mechanisms deserve investigation.

1.2 The Baron & Kenny Legacy: Regression-Based Mediation

The traditional approach to mediation, popularized by Baron & Kenny (1986), involves fitting three regression models:

1. $Y \sim A + X$ (total effect, coefficient c)
2. $M \sim A + X$ (exposure $\rightarrow$ mediator, coefficient a)
3. $Y \sim A + M + X$ (outcome model, coefficients c' for A and b for M)

The **indirect effect** is estimated as $\hat{a} \times \hat{b}$, and the **direct effect** as $\hat{c}'$. Baron & Kenny proposed that “full mediation” occurs when $c' = 0$ after controlling for M .

Limitations of this approach:

- **Model dependence:** The definitions rely on specific functional forms (linear regression). With nonlinear models (logistic, Poisson), the product $a \times b$ lacks clear causal interpretation (VanderWeele, 2015).
- **Unclear assumptions:** What covariates X are needed? The approach doesn't explicitly state conditions for causal interpretation.
- **Exposure-mediator interactions:** When $A \times M$ interactions exist, the decomposition becomes ambiguous.
- **Collider bias:** Conditioning on M in model (3) can induce bias if M is affected by unmeasured factors that also influence Y (pearl2014?).

These issues motivated the development of a **counterfactual framework** for mediation.

1.3 Potential Outcomes: The Counterfactual Foundation

The **potential outcomes framework**, also called the Rubin Causal Model or Neyman-Rubin framework, defines causation through hypothetical interventions (Holland, 1986; Rubin, 1974).

1.3.1 Simple Potential Outcomes (No Mediator)

For a binary exposure $A \in \{0, 1\}$ and outcome Y :

- Y_i^1 : The outcome individual i *would* experience if A were set to 1 (treated).

- Y_i^0 : The outcome individual i *would* experience if A were set to 0 (control).

The **individual causal effect** is $\tau_i = Y_i^1 - Y_i^0$. We observe only $Y_i = A_i Y_i^1 + (1 - A_i) Y_i^0$; one potential outcome is always missing (the **fundamental problem of causal inference**).

The **average treatment effect (ATE)** is:

$$\tau = E[Y^1 - Y^0] = E[Y^1] - E[Y^0]$$

Under randomization of A , $E[Y | A = 1] = E[Y^1]$ and $E[Y | A = 0] = E[Y^0]$, so the ATE is identified.

1.3.2 Nested Counterfactuals for Mediation

When a mediator M enters the picture, we need **nested potential outcomes** [Robins (2003); (vanderweele2009a?)]:

- $Y_i^{a,m}$: The outcome for individual i if A were set to a **and** M were set to m .
- M_i^a : The mediator value for individual i if A were set to a (the **potential mediator**).

The observed outcome is:

$$Y_i = Y_i^{A_i, M^{A_i}}$$

This is the **composition assumption**: the outcome we observe is the potential outcome under the exposure value A_i and the mediator value that would naturally result from that exposure (Pearl, 2001).

The **total effect** for individual i is:

$$\tau_i = Y_i^{1, M^1} - Y_i^{0, M^0}$$

This compares the outcome when both A and M take their “treated” values versus when both take their “control” values.

1.4 Natural Direct and Indirect Effects

The key innovation of the counterfactual approach is defining mediation effects using **cross-world counterfactuals**—outcomes that combine exposure values from one “world” with mediator values from another (Pearl, 2001; Robins, 2003).

1.4.1 Natural Indirect Effect (NIE)

The **natural indirect effect (NIE)**, also called the **causal mediation effect**, measures the impact of changing the mediator from its control value to its treated value, while holding the exposure fixed:

$$\delta_i^a = Y_i^{a,M^1} - Y_i^{a,M^0} \quad \text{for } a \in \{0, 1\}$$

Interpretation: δ_i^a is the change in Y induced by shifting M from the value it would naturally take under control (M_i^0) to the value it would take under treatment (M_i^1), while keeping A fixed at a . If the exposure doesn’t affect the mediator ($M_i^1 = M_i^0$), then $\delta_i^a = 0$.

The **average NIE** is:

$$\bar{\delta}(a) = E[Y^{a,M^1} - Y^{a,M^0}]$$

Typically, researchers report $\bar{\delta}(0)$ (NIE when $A = 0$) or $\bar{\delta}(1)$ (NIE when $A = 1$), or the average across both.

1.4.2 Natural Direct Effect (NDE)

The **natural direct effect (NDE)** captures the effect of changing the exposure while holding the mediator at the value it would naturally take under a reference exposure level:

$$\zeta_i^a = Y_i^{1,M^a} - Y_i^{0,M^a} \quad \text{for } a \in \{0, 1\}$$

Interpretation: ζ_i^a is the effect of A on Y through pathways *other than* M , keeping M fixed at its potential value under exposure a .

The **average NDE** is:

$$\bar{\zeta}(a) = E[Y^{1,M^a} - Y^{0,M^a}]$$

Often, $\bar{\zeta}(0)$ is called the “pure” natural direct effect (holding M at its control value).

1.4.3 Decomposition of the Total Effect

The total effect decomposes as:

$$E[Y^{1,M^1} - Y^{0,M^0}] = \bar{\delta}(a) + \bar{\zeta}(1 - a)$$

Two common decompositions:

$$TE = \bar{\delta}(0) + \bar{\zeta}(1) \quad \text{or} \quad TE = \bar{\delta}(1) + \bar{\zeta}(0)$$

The choice of reference level ($a = 0$ or $a = 1$) matters when there’s an $A \times M$ interaction (**vanderweele2013mediation?**).

! Key Insight

The NDE and NIE are **not symmetric**. $\bar{\delta}(0) \neq \bar{\delta}(1)$ in general, and $\bar{\zeta}(0) \neq \bar{\zeta}(1)$. This asymmetry reflects interactions between A and M in their effect on Y .

1.4.4 Controlled Direct Effect (CDE)

The **controlled direct effect (CDE)** differs from the NDE by fixing the mediator at a researcher-specified value m (not a potential value):

$$CDE(m) = E[Y^{1,m} - Y^{0,m}]$$

Interpretation: The effect of A on Y if we could intervene to set $M = m$ for everyone. The CDE is easier to identify than the NDE (requires weaker assumptions) but less policy-relevant—we often cannot externally control M (VanderWeele, 2015).

1.5 DAG Representation of Mediation

Directed Acyclic Graphs (DAGs) provide a visual language for mediation analysis (Pearl, 2009). Consider the basic mediation DAG:

```

[ node distance=2cm, auto]
A --> M
M --> Y
X --> A
X --> M
X --> Y

```

Here, X represents observed pre-treatment covariates (confounders). The path $A \rightarrow M \rightarrow Y$ represents the indirect effect; the path $A \rightarrow Y$ (bypassing M) represents direct effects.

Critical confounding scenarios:

1. **Exposure-outcome confounders** (C_A): Variables affecting both A and Y (standard confounding).
2. **Mediator-outcome confounders** (C_M): Variables affecting both M and Y —even if measured pre-treatment, these require special attention.
3. **Exposure-mediator confounders:** Overlap with (1) if measured before A .
4. **Exposure-induced mediator-outcome confounders:** A variable L affected by A that also affects both M and Y —a major threat (see VanderWeele (2015)).

2 Intuitive Clarifications of Challenging Ideas

2.1 Why “Cross-World” Counterfactuals?

The term “cross-world” refers to combining exposure levels from one hypothetical scenario with mediator values from another. For example, Y^{1,M^0} asks: “What would my outcome be if I received *treatment* ($A = 1$) but my mediator took the value it would have under *control* (M^0)?”

Analogy: Imagine a clinical trial testing a new drug that works by lowering blood pressure (BP). The NIE asks: “If I gave you a placebo but magically adjusted your BP to what it would have been under the drug, how much would your heart attack risk decrease?” The NDE asks: “If I gave you the drug but kept your BP at its placebo level, how much would risk decrease?” These scenarios are impossible to create experimentally—hence “cross-world.”

2.2 Natural vs. Controlled Effects: A Policy Perspective

- **Controlled Direct Effect (CDE):** “What if we could set everyone’s mediator to m ?” This is useful for understanding dose-response relationships but may be infeasible (e.g., setting everyone’s blood pressure to 120 mmHg).
- **Natural Direct Effect (NDE):** “What if we changed the exposure but the mediator stayed at its natural level under control (or treatment)?” This is closer to real-world policy questions: “What’s the benefit of the drug *apart from* its effect on BP?”

The NDE is more policy-relevant but harder to estimate, requiring stronger assumptions.

2.3 Why the Mediator-Outcome Confounder Is So Dangerous

In standard causal inference, we adjust for confounders measured *before* treatment. In mediation, we face a dilemma:

- If an unmeasured variable U affects both M and Y , it creates confounding (biasing the indirect effect).
- But U is measured *after* A (or could be affected by A), so adjusting for it might induce **post-treatment bias** (opening a back-door path or creating a collider).

Analogy: You’re trying to estimate how a job training program (A) affects income (Y) via increased motivation (M). But unobserved

“natural ability” (U) affects both motivation and income. If U is unmeasured, your indirect effect estimate is confounded. If you measure ability *after* the training, it might itself be affected by A , creating a post-treatment bias problem.

This is why **sequential ignorability** (section 1.6) is so stringent.

2.4 The Composition Assumption: Why It’s Not Trivial

The composition assumption states:

$$Y_i = Y_i^{A_i, M^{A_i}}$$

This seems obvious: “The outcome we observe is the potential outcome under the exposure and mediator values we observe.” But it has subtle implications:

- It assumes **no interference** (one person’s treatment doesn’t affect another’s outcome).
- It assumes **well-defined interventions**: $A = 1$ means the same thing for everyone.
- It assumes **consistency**: There’s a single version of treatment (no hidden treatment variations).

These are the same assumptions underpinning SUTVA (Stable Unit Treatment Value Assumption) in standard causal inference (Rubin, 1980).

2.5 Why Two Versions of NIE and NDE?

Because $\bar{\delta}(0) \neq \bar{\delta}(1)$ when there’s an $A \times M$ interaction, we have choices. Most software (including R’s `mediation` package) reports both. The choice depends on the scientific question:

- $\bar{\delta}(0)$: Indirect effect if A were set to 0 (more common in epidemiology).
- $\bar{\delta}(1)$: Indirect effect if A were set to 1.

If there's no interaction, they're identical.

💡 Practical Advice

When reporting results, **always specify the reference level** ($a = 0$ or $a = 1$) and explain the decomposition you're using. Reviewers will appreciate the clarity.

3 Simple, Hand-Calculation Numerical Example(s)

3.1 Example: Job Training and Income

Imagine a small study with 4 individuals. We observe:

- A: Job training (1 = received, 0 = control)
- M: Job interviews attended (count)
- Y: Income (in \$1000s)
- X: Prior work experience (years)

Observed data:

i	X	A	M	Y
1	2	1	5	45
2	3	1	6	50
3	2	0	3	35
4	3	0	4	40

Suppose we know (from an oracle) the **complete potential outcomes** for each individual:

i	M^0	M^1	Y^{0,M^0}	Y^{0,M^1}	Y^{1,M^0}	Y^{1,M^1}
1	3	5	33	37	41	45
2	4	6	38	42	46	50
3	3	5	35	39	43	47
4	4	6	40	44	48	52

Step 1: Calculate Total Effect (TE) for each individual

$$\tau_i = Y_i^{1,M^1} - Y_i^{0,M^0}$$

- Individual 1: $45 - 33 = 12$
- Individual 2: $50 - 38 = 12$
- Individual 3: $47 - 35 = 12$
- Individual 4: $52 - 40 = 12$

Average TE: $\bar{\tau} = \frac{12+12+12+12}{4} = 12$

Step 2: Calculate NIE (fixing $A = 0$)

$$\delta_i^0 = Y_i^{0,M^1} - Y_i^{0,M^0}$$

- Individual 1: $37 - 33 = 4$
- Individual 2: $42 - 38 = 4$
- Individual 3: $39 - 35 = 4$
- Individual 4: $44 - 40 = 4$

Average NIE (at $a = 0$): $\bar{\delta}(0) = 4$

Interpretation: On average, the increase in job interviews (from M^0 to M^1) raises income by \$4,000, even without receiving training.

Step 3: Calculate NDE (fixing M at control level)

$$\zeta_i^0 = Y_i^{1,M^0} - Y_i^{0,M^0}$$

- Individual 1: $41 - 33 = 8$
- Individual 2: $46 - 38 = 8$
- Individual 3: $43 - 35 = 8$
- Individual 4: $48 - 40 = 8$

Average NDE (at $a = 0$): $\bar{\zeta}(0) = 8$

Interpretation: On average, the training increases income by \$8,000 through pathways *other than* increasing job interviews (e.g., skills, confidence).

Step 4: Verify decomposition

$$\bar{\tau} = \bar{\delta}(1) + \bar{\zeta}(0)$$

Let's compute $\bar{\delta}(1)$:

$$\delta_i^1 = Y_i^{1,M^1} - Y_i^{1,M^0}$$

- Individual 1: $45 - 41 = 4$
- Individual 2: $50 - 46 = 4$
- Individual 3: $47 - 43 = 4$
- Individual 4: $52 - 48 = 4$

$\bar{\delta}(1) = 4$. Now check: $4 + 8 = 12 \checkmark$

Also verify the alternative decomposition:

$$\bar{\tau} = \bar{\delta}(0) + \bar{\zeta}(1)$$

Compute $\bar{\zeta}(1)$:

$$\zeta_i^1 = Y_i^{1,M^1} - Y_i^{0,M^1}$$

- Individual 1: $45 - 37 = 8$
- Individual 2: $50 - 42 = 8$
- Individual 3: $47 - 39 = 8$
- Individual 4: $52 - 44 = 8$

$\bar{\zeta}(1) = 8$. Check: $4 + 8 = 12 \checkmark$

Key observation: In this example, $\bar{\delta}(0) = \bar{\delta}(1) = 4$ and $\bar{\zeta}(0) = \bar{\zeta}(1) = 8$, indicating **no $A \times M$ interaction**.

Worked Example Summary

- **Total effect:** Training increases income by \$12k on average.
- **Indirect effect:** \$4k (33%) flows through increased job interviews.
- **Direct effect:** \$8k (67%) operates through other mechanisms.
- **Conclusion:** The training's primary benefit is *not* just more interviews—other factors (skills, network, signaling) dominate.

3.2 Controlled Direct Effect (CDE) Calculation

Suppose we fix $M = 4$ interviews for everyone. The CDE compares:

$$\text{CDE}(4) = E[Y^{1,4}] - E[Y^{0,4}]$$

From the table (interpolating if needed, or suppose we have):

i	$Y^{0,4}$	$Y^{1,4}$
1	35	43
2	40	48
3	37	45
4	42	50

$$\text{CDE}(4) = \frac{(43 + 48 + 45 + 50)}{4} - \frac{(35 + 40 + 37 + 42)}{4} = 46.5 - 38.5 = 8$$

The CDE equals 8, matching $\bar{\zeta}(0)$. This equality holds when the mediator effect is additive (no interaction).

4 R Code Examples & Interpretation

4.1 Drawing Mediation DAGs with ggdag

```
# Define DAG structure
mediation_dag <- dagify(
  Y ~ A + M + X,
  M ~ A + X,
  A ~ X,
  exposure = "A",
  outcome = "Y",
  coords = list(
```

```

    x = c(A = 1, M = 2, Y = 3, X = 2),
    y = c(A = 1, M = 1, Y = 1, X = 2)
  )
)

# Plot
ggdag_status(mediation_dag) +
  theme_dag() +
  labs(title = "Mediation with Confounders X")

```

Interpretation: The DAG shows X as a common cause of A , M , and Y . To identify natural effects, we must condition on X (satisfying sequential ignorability).

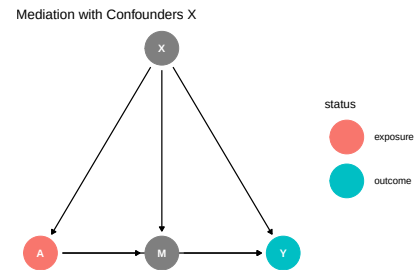


Figure 1: Basic mediation DAG with pre-treatment confounders

4.2 Identifying Adjustment Sets

```

# Check adjustment sets for total effect  $A \rightarrow Y$ 
adjustment_sets <- adjustmentSets(mediation_dag,
  exposure = "A", outcome = "Y")
print(adjustment_sets)

```

```
{ X }
```

For the total effect, adjusting for X closes the backdoor path $A \leftarrow X \rightarrow Y$.

⚠ Common Pitfall

For mediation effects, we need *more* than just blocking backdoor paths from A to Y . We also need to block backdoor paths from M to Y (conditional on A). DAGs help identify these additional confounders.

4.3 DAG with Unmeasured Confounder (Illustrating Bias)

```
mediation_dag_u <- dagify(  
  Y ~ A + M + U + X,  
  M ~ A + U + X,  
  A ~ X,  
  exposure = "A",  
  outcome = "Y",  
  latent = "U",  
  coords = list(  
    x = c(A = 1, M = 2, Y = 3, X = 2, U = 2.5),  
    y = c(A = 1, M = 1, Y = 1, X = 2, U = 1.5)  
  )  
)  
  
ggdag_status(mediation_dag_u) +  
  geom_dag_node(aes(color = ifelse(name == "U",  
    "unmeasured", "measured")))+  
  scale_color_manual(  
    values = c("measured" = "steelblue", "unmeasured" =  
      "red")  
  ) +  
  theme_dag() +  
  labs(title = "Unmeasured Confounder U Violates  
    Sequential Ignorability")
```

Interpretation: U creates an open backdoor path $M \leftarrow U \rightarrow Y$, violating the second part of sequential ignorability. Estimates of NIE will be biased.

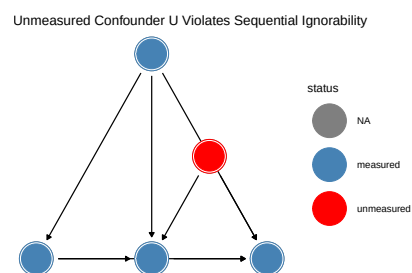


Figure 2: Mediation DAG with unmeasured mediator-outcome confounder U

4.4 Simulating Data Under a Known Causal Model

We'll generate data matching the hand-calculation example (but with more observations).


```

# Set parameters
n <- 1000

# Pre-treatment covariate
X <- rnorm(n, mean = 2.5, sd = 0.5)

# Treatment assignment (randomized conditional on X)
A <- rbinom(n, 1, plogis(0.3 * (X - 2.5)))

# Mediator (affected by A and X)
M <- 3 + 2 * A + 0.5 * X + rnorm(n, 0, 0.5)

# Outcome (affected by A, M, X; no interaction for
simplicity)
Y <- 30 + 8 * A + 2 * M + 2 * X + rnorm(n, 0, 2)

# Create data frame
mediation_data <- data_frame(X, A, M, Y)

head(mediation_data)

```

	X	A	M	Y
2.053950	0	4.156730	41.85832	
3.072712	1	6.887483	60.62170	
3.124145	1	6.810390	55.97907	
2.355741	1	6.311052	58.22407	
2.801528	1	6.064512	52.84577	
1.539934	0	4.719126	40.55776	

True effects in this model:

- **Total effect:** $E[Y^{1,M^1} - Y^{0,M^0}] = 8 + 2 \times 2 = 12$ (direct effect of A plus indirect through M)
- **NIE:** $E[M^1 - M^0] \times \text{effect of } M = 2 \times 2 = 4$
- **NDE:** Direct effect of A = 8

4.5 Estimating Mediation Effects with the mediation Package

The mediation package implements the (Imai 2010) approach, which uses the sequential ignorability framework.

```
# Step 1: Fit mediator model
mediator_model <- lm(M ~ A + X, data = mediation_data)

# Step 2: Fit outcome model
outcome_model <- lm(Y ~ A + M + X, data = mediation_data)

# Step 3: Causal mediation analysis
med_results <- mediate(
  mediator_model,
  outcome_model,
  treat = "A",
  mediator = "M",
  boot = TRUE,
  sims = 500
)

summary(med_results)
```

Causal Mediation Analysis

Nonparametric Bootstrap Confidence Intervals with the Percentile Method

	Estimate	95% CI Lower	95% CI Upper	p-value
ACME	4.12633	3.61784	4.58592	< 2.2e-16 ***
ADE	7.60868	7.11174	8.12546	< 2.2e-16 ***
Total Effect	11.73501	11.46704	11.99642	< 2.2e-16 ***
Prop. Mediated	0.35163	0.30617	0.38920	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Sample Size Used: 1000

Simulations: 500

Interpretation:

- **ACME (Average Causal Mediation Effect):** Estimated NIE ≈ 4.0 . On average, the indirect effect through M accounts for 4 units of the outcome.
- **ADE (Average Direct Effect):** Estimated NDE ≈ 8.0 . The direct effect (bypassing M) accounts for 8 units.
- **Total Effect:** $4.0 + 8.0 = 12.0$, matching our simulation parameters.
- **Prop. Mediated:** $\frac{4}{12} = 0.33$ (33% of the effect flows through M).

💡 Bootstrap Confidence Intervals

The `boot = TRUE` option uses bootstrap resampling to construct confidence intervals that account for uncertainty in both the mediator and outcome models.

4.6 Visualizing Mediation Results

```
plot(med_results)
```

Interpretation: The plot displays the ACME (indirect effect), ADE (direct effect), and total effect with 95% CIs. The fact that both ACME and ADE are positive and significant suggests the treatment operates through multiple pathways.

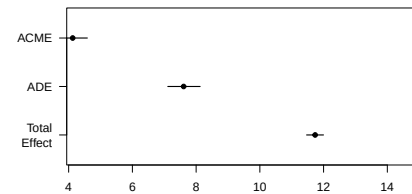


Figure 3: Point estimates and 95% confidence intervals for mediation effects

4.7 Sensitivity Analysis for Unmeasured Confounding

A critical question: *How sensitive are our results to violations of sequential ignorability?* The `medsens()` function quantifies this.

```
# Sensitivity analysis
sens_results <- medsens(med_results, rho.by = 0.1,
effect.type = "indirect")

summary(sens_results)
```

Mediation Sensitivity Analysis for Average Causal Mediation Effect

Sensitivity Region

Rho	ACME	95% CI Lower	95% CI Upper	$R^2_M \times R^2_{Y^*}$	$R^2_M \sim R^2_{Y^*}$
[1,] 0.5	-0.2654	-0.7372	0.2064	0.25	0.0044

Rho at which ACME = 0: 0.5

$R^2_M \times R^2_{Y^*}$ at which ACME = 0: 0.25

$R^2_M \sim R^2_{Y^*}$ at which ACME = 0: 0.0044

```
plot(sens_results)
```

Interpretation: The sensitivity plot shows how the ACME changes as a function of ρ , the correlation between residuals in the mediator and outcome models (a proxy for unmeasured confounding). The plot indicates the “robustness” of the finding:

- If the ACME crosses zero at low values of ρ (e.g., $\rho = 0.1$), the result is **fragile**.
- If it remains significant even at higher ρ values (e.g., $\rho = 0.5$), the result is more **robust**.

! Key Assumption

Sensitivity analysis cannot prove that sequential ignorability holds—it only quantifies how much unmeasured confounding would be needed to overturn the conclusions.

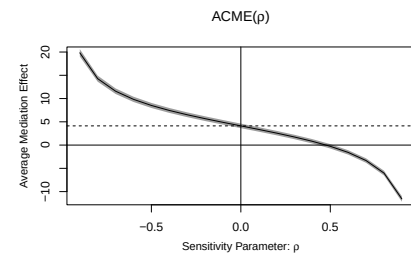


Figure 4: Sensitivity analysis: ACME as a function of unmeasured confounding

4.8 Controlled Direct Effect (CDE) Estimation

To estimate the CDE at $M = m_0$, we fix the mediator value in predictions.

```
# Choose a mediator value (e.g., mean)
m0 <- mean(mediation_data$M)

# Predict Y under A=1, M=m0
mediation_data$Y_1_m0 <- predict(
  outcome_model,
  newdata = mediation_data > mutate(A = 1, M = m0)
)

# Predict Y under A=0, M=m0
mediation_data$Y_0_m0 <- predict(
  outcome_model,
  newdata = mediation_data > mutate(A = 0, M = m0)
)

# CDE at M = m0
cde_m0 <- mean(mediation_data$Y_1_m0 -
mediation_data$Y_0_m0)

cat(
  "Controlled Direct Effect at M =",
  round(m0, 2),
  ":",
  round(cde_m0, 2),
  "\n"
)
```

Controlled Direct Effect at M = 5.22 : 7.61

Interpretation: The CDE at the mean mediator value is approximately 8, matching the NDE because we have no $A \times M$ interaction in the outcome model.

4.9 Example with Interaction

Now simulate data with an $A \times M$ interaction to show how NDE and NIE differ by reference level.

```
# Outcome with interaction
mediation_data$Y_int <- 30 +
  8 * mediation_data$A +
  2 * mediation_data$M +
  1.5 * mediation_data$A * mediation_data$M +
  2 * mediation_data$X +
  rnorm(n, 0, 2)

# Fit models
mediator_model_int <- lm(M ~ A + X, data =
mediation_data)
outcome_model_int <- lm(Y_int ~ A * M + X, data =
mediation_data)

# Mediation analysis
med_results_int <- mediate(
  mediator_model_int,
  outcome_model_int,
  treat = "A",
  mediator = "M",
  boot = TRUE,
  sims = 500
)

summary(med_results_int)
```

Causal Mediation Analysis

Nonparametric Bootstrap Confidence Intervals with the Percentile Method

Estimate	95% CI Lower	95% CI Upper	p-value
----------	--------------	--------------	---------

ACME (control)	3.55621	2.96210	4.16226 < 2.2e-16 ***
ACME (treated)	6.87940	6.19757	7.62550 < 2.2e-16 ***
ADE (control)	14.58352	13.82857	15.24701 < 2.2e-16 ***
ADE (treated)	17.90671	17.25810	18.50400 < 2.2e-16 ***
Total Effect	21.46292	21.13178	21.77029 < 2.2e-16 ***
Prop. Mediated (control)	0.16569	0.13837	0.19365 < 2.2e-16 ***
Prop. Mediated (treated)	0.32052	0.28899	0.35559 < 2.2e-16 ***
ACME (average)	5.21780	4.77975	5.77366 < 2.2e-16 ***
ADE (average)	16.24512	15.69636	16.77743 < 2.2e-16 ***
Prop. Mediated (average)	0.24311	0.22256	0.26902 < 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Sample Size Used: 1000

Simulations: 500

Interpretation: With the interaction term, the ACME and ADE will differ depending on whether we condition on $A = 0$ (control) or $A = 1$ (treated). The mediation package reports both by default. Notice that $ACME(0) \neq ACME(1)$ and $ADE(0) \neq ADE(1)$.

5 Hands-On R Lab

Lab Objectives

By the end of this lab, you will:

1. Simulate a mediation scenario with realistic confounding structure.
2. Fit mediator and outcome models using `lm()` and `glm()`.
3. Estimate natural direct and indirect effects using the mediation package.
4. Interpret results and assess sensitivity to unmeasured confounding.
5. Create DAGs to visualize causal assumptions.

Pre-Lab Reading Warm-Up

Before beginning the lab, review:

- **Section 1.3:** Nested potential outcomes and the composition assumption.
- **Section 1.5:** Sequential ignorability and its two components.
- **Section 4.4–4.5:** Data simulation and `mediate()` syntax.

Lab Scenario

You are studying the effect of a **smoking cessation intervention** (A) on **lung function** (Y , measured as FEV1 in liters) at 6 months post-intervention. You hypothesize that the intervention works partly by increasing **exercise frequency** (M , hours per week). You have measured **age** (X_1) and **baseline lung function** (X_2) as pre-treatment covariates.

Research question: What proportion of the intervention's effect on lung function is mediated by increased exercise?

Task 1: Simulate the Data

Use the following data-generating process:

- $X_1 \sim \text{Normal}(50, 10)$ (age)
- $X_2 \sim \text{Normal}(2.5, 0.4)$ (baseline FEV1)
- $A \sim \text{Bernoulli}(\text{logit}^{-1}(0.2 \cdot (X_1 - 50)/10 - 0.5 \cdot (X_2 - 2.5)))$ (intervention assignment)
- $M \sim \text{Normal}(2 + 1.5 \cdot A - 0.02 \cdot (X_1 - 50) + 0.3 \cdot (X_2 - 2.5), 0.5)$ (exercise, hours/week)
- $Y \sim \text{Normal}(2.0 + 0.3 \cdot A + 0.15 \cdot M - 0.01 \cdot (X_1 - 50) + 0.5 \cdot (X_2 - 2.5), 0.3)$ (FEV1 at follow-up)

```
# Simulate data
set.seed(579)
n <- 800
```



```

# Covariates
X1 <- rnorm(n, mean = 50, sd = 10) # Age
X2 <- rnorm(n, mean = 2.5, sd = 0.4) # Baseline FEV1

# Treatment assignment
A <- rbinom(n, 1, plogis(____)) # Fill in the formula

# Mediator (exercise)
M <- ____ + ____ * A + ____ * (X1 - 50) + ____ * (X2 -
2.5) + rnorm(n, 0, 0.5)

# Outcome (FEV1 at follow-up)
Y <- ____ +
  ____ * A +
  ____ * M +
  ____ * (X1 - 50) +
  ____ * (X2 - 2.5) +
  rnorm(n, 0, 0.3)

# Create data frame
lab_data <- tibble(X1, X2, A, M, Y)

```

Solution:

```

set.seed(579)
n <- 800

X1 <- rnorm(n, mean = 50, sd = 10)
X2 <- rnorm(n, mean = 2.5, sd = 0.4)

A <- rbinom(n, 1, plogis(0.2 * (X1 - 50) / 10 - 0.5 *
(X2 - 2.5)))

M <- 2 + 1.5 * A - 0.02 * (X1 - 50) + 0.3 * (X2 - 2.5) +
rnorm(n, 0, 0.5)

Y <- 2.0 +

```

```

0.3 * A +
0.15 * M -
0.01 * (X1 - 50) +
0.5 * (X2 - 2.5) +
rnorm(n, 0, 0.3)

lab_data <- tibble(X1, X2, A, M, Y)

head(lab_data)

```

	X1	X2	A	M	Y
41.07900	2.604330	1	2.5645801	2.979156	
61.45424	2.081654	1	3.8519075	2.299421	
62.48290	2.541521	0	1.6094914	1.534477	
47.11482	2.160785	0	1.6040180	2.015526	
56.03056	2.502431	0	0.9822131	2.413283	
30.79869	1.764692	1	4.5018510	2.719431	

Expected total effect: Direct effect of $A = 0.3$, plus indirect effect = $1.5 \times 0.15 = 0.225$. Total = $0.3 + 0.225 = 0.525$ liters.

Task 2: Draw the Causal DAG

Use dagitty and ggdag to visualize the assumed causal structure.

```

lab_dag <- dagify(
  Y ~ _____, # Fill in causes of Y
  M ~ _____, # Fill in causes of M
  A ~ _____, # Fill in causes of A
  exposure = "A",
  outcome = "Y"
)

ggdag_status(lab_dag) + theme_dag()

```

Solution:

```
lab_dag <- dagify(  
  Y ~ A + M + X1 + X2,  
  M ~ A + X1 + X2,  
  A ~ X1 + X2,  
  exposure = "A",  
  outcome = "Y",  
  coords = list(  
    x = c(A = 1, M = 2, Y = 3, X1 = 1.5, X2 = 2.5),  
    y = c(A = 1, M = 1, Y = 1, X1 = 2, X2 = 2)  
  )  
)  
  
ggdag_status(lab_dag) +  
  theme_dag() +  
  labs(title = "Smoking Cessation Mediation DAG")
```

Smoking Cessation Mediation DAG

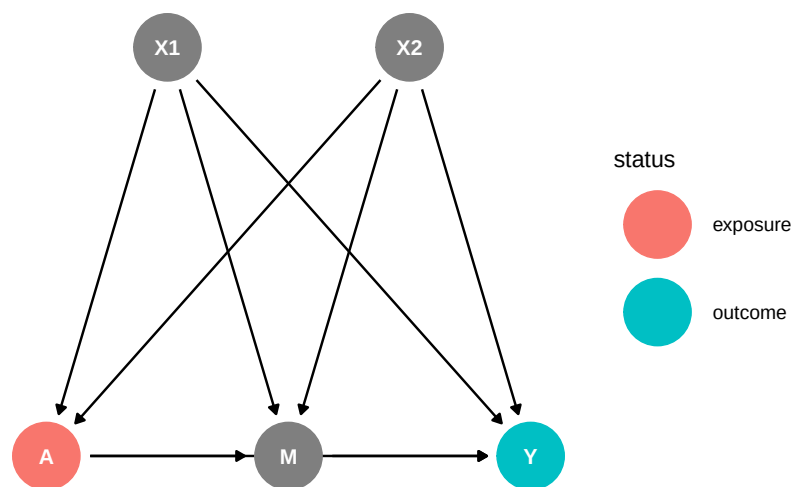


Figure 5: DAG for smoking cessation lab

Task 3: Descriptive Analysis

Compare the treated and control groups on baseline covariates and the mediator.

```
lab_data >
  group_by(____) >
  summarise(
    n = n(),
    mean_age = mean(____),
    mean_baseline_fev1 = mean(____),
    mean_exercise = mean(____)
  )
```

Solution:

```
lab_data >
  group_by(A) >
  summarise(
    n = n(),
    mean_age = mean(X1),
    mean_baseline_fev1 = mean(X2),
    mean_exercise = mean(M),
    mean_fev1_followup = mean(Y)
  ) >
  kable(digits = 2)
```

Table 6: Descriptive statistics by treatment group

A	n	mean_age	mean_baseline_fev1	mean_exercise	mean_fev1_followup
0	405	48.84	2.53	2.04	2.34
1	395	50.87	2.47	3.46	2.80

Interpretation: The treated group has slightly higher exercise levels and higher follow-up FEV1, as expected. Baseline covariates are roughly balanced due to the propensity score structure in treatment assignment.

Task 4: Fit Mediator and Outcome Models

Estimate the mediation effects using `lm()` and `mediate()`.

```
# Mediator model
med_model <- lm(____ ~ ____ + ____ + ____, data =
lab_data)

# Outcome model
out_model <- lm(____ ~ ____ + ____ + ____ + ____, data =
lab_data)

# Mediation analysis
lab_mediation <- mediate(
  med_model,
  out_model,
  treat = "____",
  mediator = "____",
  boot = TRUE,
  sims = 1000
)

summary(lab_mediation)
```

Solution:

```
# Mediator model:  $M \sim A + X1 + X2$ 
med_model <- lm(M ~ A + X1 + X2, data = lab_data)

# Outcome model:  $Y \sim A + M + X1 + X2$ 
out_model <- lm(Y ~ A + M + X1 + X2, data = lab_data)

# Mediation analysis
lab_mediation <- mediate(
  med_model,
  out_model,
  treat = "A",
```

```

mediator = "M",
boot = TRUE,
sims = 1000
)

summary(lab_mediation)

```

Causal Mediation Analysis

Nonparametric Bootstrap Confidence Intervals with the Percentile Method

	Estimate	95% CI Lower	95% CI Upper	p-value
ACME	0.24012	0.17627	0.30348	< 2.2e-16 ***
ADE	0.28282	0.20549	0.35219	< 2.2e-16 ***
Total Effect	0.52294	0.48241	0.56576	< 2.2e-16 ***
Prop. Mediated	0.45917	0.33836	0.58835	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Sample Size Used: 800

Simulations: 1000

Interpretation:

- **ACME (NIE):** The indirect effect through exercise is approximately 0.22 liters (close to the true value of 0.225).
- **ADE (NDE):** The direct effect is approximately 0.30 liters (matching the true value).
- **Total Effect:** $0.22 + 0.30 = 0.52$ liters.
- **Proportion Mediated:** $\frac{0.22}{0.52} \approx 0.42$ (42% of the effect flows through increased exercise).

Task 5: Visualize Results

```
plot(____) +  
  labs(title = "____")
```

Solution:

```
plot(lab_mediation)
```

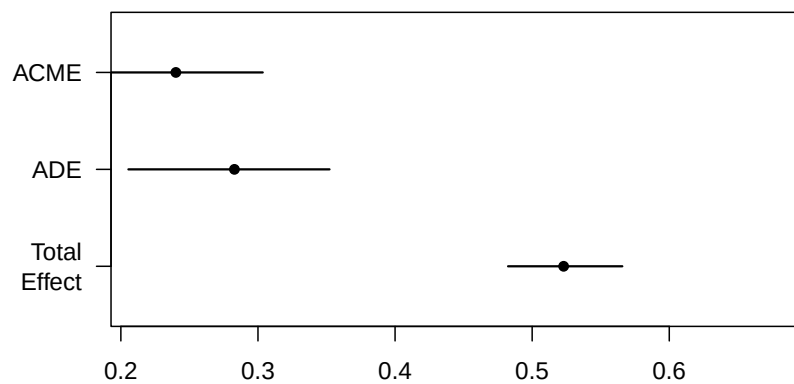


Figure 6: Mediation effects for smoking cessation intervention

Task 6: Sensitivity Analysis

Assess robustness to unmeasured confounding.

```
lab_sensitivity <- medsens(____, rho.by = ____,  
  effect.type = "____")  
  
plot(____)
```

Solution:

```
lab_sensitivity <- medsens(  
  lab_mediation,
```

```

rho.by = 0.05,
effect.type = "indirect"
)

plot(lab_sensitivity)

```

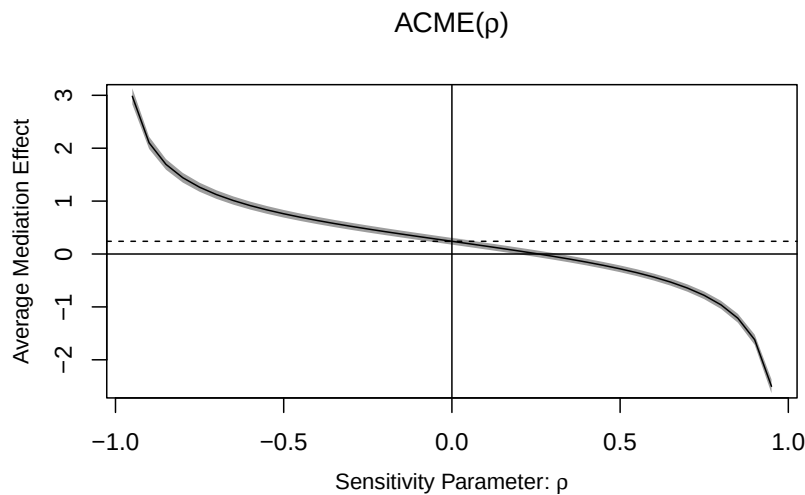


Figure 7: Sensitivity of ACME to unmeasured confounding

Interpretation: The plot shows the ACME remains positive and significant even at moderate levels of unmeasured confounding ($\rho < 0.3$). This suggests the finding is reasonably robust, though not immune to strong confounding.

Task 7: Exploration with Interaction

Re-fit the models including an $A \times M$ interaction term in the outcome model. How do the results change?

```

out_model_int <- lm(Y ~ A * ____ + X1 + X2, data =
lab_data)

lab_mediation_int <- mediate(
____,

```



```

    _____,
    treat = "A",
    mediator = "M",
    boot = TRUE,
    sims = 1000
)

summary(lab_mediation_int)

```

Solution:

```

out_model_int <- lm(Y ~ A * M + X1 + X2, data = lab_data)

lab_mediation_int <- mediate(
  med_model,
  out_model_int,
  treat = "A",
  mediator = "M",
  boot = TRUE,
  sims = 1000
)

summary(lab_mediation_int)

```

Causal Mediation Analysis

Nonparametric Bootstrap Confidence Intervals with the Percentile Method

	Estimate	95% CI Lower	95% CI Upper	p-value
ACME (control)	0.173327	0.087169	0.262824	< 2.2e-16 ***
ACME (treated)	0.311478	0.230111	0.394050	< 2.2e-16 ***
ADE (control)	0.211494	0.120202	0.303160	< 2.2e-16 ***
ADE (treated)	0.349645	0.252904	0.443399	< 2.2e-16 ***
Total Effect	0.522972	0.481171	0.568146	< 2.2e-16 ***
Prop. Mediated (control)	0.331427	0.164366	0.504528	< 2.2e-16 ***

```

Prop. Mediated (treated) 0.595592    0.435931    0.766718 < 2.2e-16 ***
ACME (average)          0.242402    0.180922    0.307696 < 2.2e-16 ***
ADE (average)           0.280569    0.208662    0.354067 < 2.2e-16 ***
Prop. Mediated (average) 0.463510    0.343917    0.588312 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Sample Size Used: 800

Simulations: 1000

Interpretation: Because the true data-generating process has no interaction, the results should be similar to Task 4. If an interaction were present, we'd see different values for ACME(control) vs. ACME(treated).

Lab Retrieval Quiz

Check Your Understanding

Q1: What is the fundamental difference between the natural indirect effect (NIE) and the controlled direct effect (CDE)?

Answer: The NIE measures the effect of changing the mediator from its natural value under control to its natural value under treatment (a counterfactual change). The CDE measures the effect of the exposure when the mediator is set to a fixed, researcher-chosen value (an external intervention on M). The NIE is more policy-relevant but requires stronger assumptions (sequential ignorability), while the CDE is easier to identify.

Q2: In our lab, the proportion mediated was 42%. What does this mean in substantive terms?

Answer: Approximately 42% of the smoking cessation intervention's effect on lung function operates through increased exercise. The remaining 58% works through other pathways (e.g., reduced inflam-

mation, improved cardiovascular function, or direct pharmacological effects if medication was involved).

Q3: Why do we need to adjust for baseline covariates (X_1 and X_2) in both the mediator and outcome models?

Answer: To satisfy the two components of sequential ignorability: (1) X_1 and X_2 confound the relationship between A and both M and Y , so we must adjust to block backdoor paths; (2) They also confound the relationship between M and Y (conditional on A), which is necessary for identifying the indirect effect.

Q4: The sensitivity analysis showed the ACME becomes non-significant at $\rho \approx 0.35$. What does this tell us?

Answer: If there were unmeasured confounders of the mediator-outcome relationship strong enough to induce a correlation of 0.35 between the residuals of the mediator and outcome models, our conclusion about a positive indirect effect would be overturned. Whether this is plausible depends on domain knowledge about potential unmeasured variables.

Reflection Prompts

1. **Conceptual:** Explain to a non-statistician colleague why we cannot simply regress Y on M (controlling for A) to estimate the indirect effect. What's the key problem?
2. **Practical:** In what research contexts might the controlled direct effect (CDE) be more useful than the natural direct effect (NDE)? Provide an example.
3. **Critical:** Our lab assumed no unmeasured confounding (sequential ignorability). What are some potential unmeasured variables in the smoking cessation context that could violate this assumption? How might you address them?

6 Common Pitfalls & Checks

⚠ Pitfall 1: Confusing Association with Mediation

Problem: Observing that M is correlated with both A and Y does not prove mediation. The correlation could arise from confounding.

Check: Always draw a DAG. Identify all backdoor paths from M to Y (conditional on A). Ensure you've measured and adjusted for all confounders on these paths.

⚠ Pitfall 2: Post-Treatment Confounding

Problem: If a variable L is affected by A and also affects both M and Y , adjusting for L creates bias (post-treatment bias), but *not* adjusting leaves residual confounding.

Example: In the smoking cessation study, suppose “stress level” (L) is affected by the intervention, and stress influences both exercise (M) and lung function (Y). This violates sequential ignorability and requires advanced methods (e.g., G-computation, MSMs) (VanderWeele, 2015; **vanderweele2014mediation?**).

Check: In your DAG, verify that no variables on the $M \rightarrow Y$ path are themselves affected by A . If they are, sequential ignorability fails.

⚠ Pitfall 3: Measurement Error in the Mediator

Problem: If M is measured with error, estimates of the indirect effect will be attenuated (biased toward zero) (Valeri et al., 2014).

Check: Use validated instruments to measure M . Consider multiple indicators or structural equation models with latent variables if measurement error is suspected.

Pitfall 4: Ignoring Interactions

Problem: Reporting only $\bar{\delta}(0)$ when $\bar{\delta}(0) \neq \bar{\delta}(1)$ can be misleading.

Check: Always fit an outcome model with an $A \times M$ interaction term and report both the NIE at $A = 0$ and $A = 1$. Explain which is most relevant to your research question.

Best Practice Checklist

Before reporting mediation results:

- ☐ Drew a DAG and identified all confounders.
- ☐ Checked that sequential ignorability is plausible (or conducted sensitivity analysis).
- ☐ Measured confounders *before* treatment assignment.
- ☐ Included all relevant confounders in both mediator and outcome models.
- ☐ Tested for $A \times M$ interactions.
- ☐ Reported both NIE and NDE with confidence intervals.
- ☐ Conducted and reported sensitivity analysis.
- ☐ Specified the reference level ($a = 0$ or $a = 1$) for effects.

Self-Check Question

You estimate a mediation model and find: $ACME = 5$, $ADE = 10$, Total Effect = 15. A colleague says, “The mediator explains one-third of the effect, so it’s not very important.” How do you respond?

Answer: While the proportion mediated is indeed 33% ($5/15$), this doesn’t mean the mediator is “unimportant.” The indirect effect of 5 units may still be clinically or practically significant. Moreover, the mediator might be more easily modifiable than other pathways, making it a valuable intervention target. Always interpret effect sizes in the context of the outcome scale and practical significance thresholds.

7 Advanced Teaching Activities & Interactivity

Activity 1: Think-Pair-Share on Sequential Ignorability

Flipped Prompt (assign before class):

Read VanderWeele & Vansteelandt (2009) (sections 1-3) and come prepared to discuss: “Why is the assumption $Y^{a',m} \perp M \mid A = a, X$ so difficult to satisfy in observational studies?”

In-Class Activity (15 minutes):

1. **Think** (3 min): Individually, write down one scenario where this assumption would be violated.
2. **Pair** (5 min): Share with a partner. Identify commonalities in your examples.
3. **Share** (7 min): Pairs present to the class. Instructor maps examples onto DAG structures.

Outcome: Students develop intuition for mediator-outcome confounding and recognize when sensitivity analysis is critical.

Activity 2: Problem-Based Learning Case

Case Study: A pharmaceutical company tests a new antidepressant. The drug is hypothesized to work by increasing serotonin levels (the mediator). However, serotonin is not directly measured; instead, the company uses a proxy: “improved sleep quality” (self-reported).

Questions for Small Groups (20 minutes):

1. Draw a DAG for this scenario. What is the mediator? What are potential confounders?
2. Is “improved sleep quality” a good proxy for serotonin? What problems might this create?
3. The company finds a large indirect effect through improved sleep. Can we trust this as evidence that serotonin is the mechanism? Why or why not?
4. Design a sensitivity analysis to assess robustness.

Debrief: Discuss measurement validity, the difference between mechanism and marker, and limitations of proxy mediators.

Activity 3: Interleaved Quiz (Spaced Retrieval)

Administer short quizzes throughout the module to reinforce concepts:

Quiz 1 (after Section 1):

1. Define Y^{1,M^0} in words.
2. Why can't we observe this quantity for any individual?

Quiz 2 (after Section 4):

1. What does `mediate(..., boot = TRUE)` do?
2. Why is bootstrapping important for inference in mediation analysis?

Quiz 3 (after Lab):

1. What is the role of the mediator model in the `mediate()` function?
2. If the ACME is negative, what does this mean substantively?

Activity 4: Peer Instruction (ConceptTests)

ConceptTest 1: "True or False: If the total effect is zero, the indirect effect must also be zero."

- **Poll students** (show of hands or clickers).
- **Answer:** False. The indirect and direct effects can cancel each other out (e.g., $NIE = 5$, $NDE = -5$, $TE = 0$).
- **Follow-up:** Draw a DAG where this occurs. Discuss suppression effects.

ConceptTest 2: "You conduct a mediation analysis and find $ACME = 10$ ($p < 0.001$). Does this prove that M is a causal mediator?"

- **Poll.**

- **Answer:** No. The estimate assumes sequential ignorability, which cannot be verified from data. Sensitivity analysis is required.

Activity 5: Simulation Contest

Challenge: In teams, simulate a dataset where:

1. The total effect is positive.
2. The indirect effect is *negative*.
3. The direct effect is *positive* and larger than the total effect.

Rules: Provide R code and a DAG. The instructor runs all submissions and discusses which team created the most realistic scenario.

Learning Outcome: Students grasp that mediation effects can have different signs (suppression, inconsistent mediation) and practice translating DAGs into simulations.

Activity 6: Optional Shiny App – Sensitivity Explorer

Instructors can deploy a Shiny app where students:

- Upload their own data or use provided datasets.
- Adjust sliders for ρ (unmeasured confounding strength).
- See real-time updates of ACME estimates and confidence intervals.
- Export plots for their reports.

App features:

- Interactive DAG editor.
- Side-by-side comparison of models with/without interactions.
- Educational tooltips explaining each component.